MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 1103 COMPUTER AND COMMUNICATION TECHNOLOGY

END OF SEMESTER EXAMINATION

DATE: JULY, 2016 TIME ALLOWED: 3 HOURS

INSTRUCTIONS

1. There are **FIVE** questions in this paper. Please check.
2. Answer **ALL** questions.
3. Marks are indicated for each question.

QUESTION ONE

(a) Define network topology. [2 Marks]

(b) Describe the **two** main parts of a computer. [4 Marks]

(c) Distinguish the following:-

(i) Storage and Memory. [2 Marks]

(ii) Data and Information. [2 Marks]

(iii) Synchronous transmission and asynchronous transmission. [2 Marks]

(iv) Field and record. [2 Marks]

(v) Data rate and bandwidth. [2 Marks]

(d) Outline any **five** input and **five** output devices of the computer. [10 Marks]

QUESTION TWO

(a) Describe any **five** advantages and **five** disadvantages of a computer. [10 Marks]

(b) Explain any **four** security threats in ICT and their remedies. [8 Marks]

(b) Explain any **two** types of information systems. [4 Marks]

(d) Describe any **two** advantages and **two** disadvantages for each of the following ways in which signed binary numbers may be expressed:

(i) Signed magnitude. [4 Marks]

(ii) Two's complement. [4 Marks]

QUESTION THREE

(a) Using signed magnitude binary arithmetic, find the sum of 75 and 46. [5 Marks]

(b) Find the sum of $CAB_{16} + BED_{16}$. [5 Marks]

(c) Convert the following binary numbers to their decimal equivalents:

(i) 10010110_2. [5 Marks]

(ii) 10000001_2. [5 Marks]

(d) Using two's complement binary arithmetic, find the sum of 107 and 46. [8 Marks]

QUESTION FOUR

(a) Explain any **three** types of transmission media. [6 Marks]

(b) Describe any **three** main types of computer networks. [6 Marks]

(c) Describe any **five** advantages and **five** disadvantages of computer networks. [10 Marks]

(d) Describe any **two** advantages and **two** disadvantages for each of following three network topologies:

(i) Ring. [4 Marks]

(ii) Bus. [4 Marks]

(iii) Star. [4 Marks]

QUESTION FIVE

(a) Define the term database. [2 Marks]

(b) Explain any **four** advantages of database management systems. [4 Marks]

(c) You have been hired to design a database for a company. Describe any **two** types of database models that you are likely to choose. [6 Marks]

(d) Describe Systems Development Life Cycle (SDLC). [12 Marks]

END OF QUESTION PAPER